Probability and Random Processes (EE2011)

Date: April 06, 2026

Course Instructor(s)

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Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 65

Total Questions: 6

$f(x) = \dfrac{\binom{3}{x}\binom{2}{y}\binom{3}{4-x-y}}{\binom{8}{4}}$

Roll No ______ Section ______ Student Signature ______

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1. **Attempt all the questions. Answers must not be given numerically only; explicitly state which probability is being calculated in each part.**

CLO # 02: Interpret the concept of random variables, probability distribution/density functions, expected values of a random variable, regression and curve fitting.

Q1. From a sack of fruit containing 3 oranges, 2 apples and 3 bananas, a random sample of 4 pieces of fruit is selected. If X is the number of oranges and Y is the number of apples in the sample, 2.5 **[5+5=10 marks]**

Determine, (a) the joint probability distribution of X and Y; [5]

(b) $P[(X,Y)\in A]$, where A is the region that is given by $\{(x,y)|x+y\leq 2\}$. $\frac{1}{2}$

Q2. *A delivery company estimates the expected delivery time for packages and denotes it by t. The actual delivery time X (in hours) has the following probability density function:*

$$f(x) = \begin{cases} \dfrac{4}{3t}, & \frac{t}{4} \leq x \leq t \\ 0, & \text{otherwise} \end{cases}$$

Determine, (a) the cumulative distribution function F(x) **[5+5=10 marks]**

(b) The probability that the actual delivery time is less than the estimated time, i.e. P(X < t). 0

$\int_{t/4}^{t} \frac{4}{3t} \rightarrow \frac{4}{3}\int \frac{1}{t}$

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Q3. The life span in hours of an electrical component is a random variable with cumulative distribution function

[5+5=10 marks]

$$F(x) = \begin{cases} 1 - e^{-\frac{x}{50}}, & x > 0, \\ 0, & \text{eleswhere.} \end{cases}$$

a. **Determine,** (a) the probability density function f(x) ✓ 5

b. The probability that the life span of such a component will exceed 70 hours. 0

Q4. In a warehouse, X and Y represent the number of boxes of two different products packed by two workers in a day. Their joint probability density function is given by

$f(x|y)$

$$f(x, y) = \begin{cases} \frac{6-x-y}{8}, & 0 < x < 2,\ 2 < y < 4, \\ 0, & \text{elsewhere,} \end{cases}$$

$P(A \cap B)$
$x = 1$
$y = 1$ mark

If the first worker packs exactly 1 box, **compute** the conditional probability that the second worker packs between 1 and 3 boxes.

[5 marks] → 3

Q5. The random variables X and Y represent the amounts of time (in hours) that two machines operate during a day in an industrial system. Their joint probability density function is given by

$$f(x, y) = \begin{cases} \frac{2}{7}(x + 2y), & 0 < x < 1,\ 1 < y < 2, \\ 0, & \text{elsewhere,} \end{cases}$$

Determine,

[5+5+5=15 marks]

a. The marginal density functions of X and Y. 5
b. The total operating time of both machines does not exceed 2 hours. 0 $P(0,1)$
c. Are X and Y independent? 2

Q6. X and Y be the random variables with joint probability distribution f(x,y). $P(1,0)$

$f(x,y)$		x = 1	2	3
	1	0.05	0.05	0.10
y	3	0.05	0.10	0.35
	5	0.00	0.20	0.10
		X	Y	Z

$\int_0^2 \int_1^2$

Calculate

[5+5+5=15 marks]

5 ← a. The expected value of $g(X, Y) = XY^2$
b. μ_X and μ_Y 0
c. The correlation coefficient of X and Y, ρ_{XY} 0

μ $(1)(X) + (2)(Y) + (3)(Z)$

Good Luck! ☺

$g(x)$

$g(1,1)$
$\forall y$ $= (1)(1)^2(0.05)$,

$g(x) =$

$(2)(5)^2(0.2)$

Fall